

Paper 2 · Answers & Solutions

2026 · 45 questions

Part 1 · Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	B	C	A	D	D	A	C	C	B	D	C	A	B	D
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	C	C	B	D	D	B	A	A	C	C	D	A	D	B
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
C	A	A	B	D	A	B	A	B	D	D	C	B	B	C

Part 2 · Worked Solutions

Section A

1 A

B-
Polynomials

- 1 Expand $(p - 2q)(p^2 + 2pq - 4q^2)$
- 2 $= p^3 - 8pq^2 + 8q^3$, choose A

2 B

B-
Laws of Indices

- 1 $(2x^3)^4 = 16x^{12}$
- 2 $\frac{16x^{12}}{8x^5} = 2x^7$, choose B

3 C

B
Simultaneous Equations

- 1 $5x - 3y = 29$, $3x + 4y = 29$
- 2 Solving gives $y = 2$, choose C

4 A

B
Identities

- 1 Expand both sides
- 2 $a = -5$, $b = -5$, choose A

5 D

B
Change of Subject

- 1 $\frac{3}{k-2} = 5-h \Rightarrow k-2 = \frac{3}{5-h}$
- 2 $k = \frac{13-2h}{5-h}$, choose D

6 D

B+
Approximation

- 1 Both bounds round to **0.0847** (3 sig. fig.)
- 2 choose D

7 A

B+
Inequalities

- 1 $x \geq -2$ or $x > 12$
- 2 Union $x \geq -2$; least integer -2 , choose A

8 C

B+
Functions

- 1 $f(c) = c$; $f(-c) = 2c^4 + c$
- 2 Sum $= 2c^4 + 2c$, choose C

9 C

B+
Factor Theorem

- 1 Divisible by $3x + k \Rightarrow$ put $x = -\frac{k}{3}$
- 2 Solving gives $k = 9$, choose C

10 B

B+
Graphs of Quadratic Functions

- 1 $y = -x^2 + 2x + 7$; opens downward (I true)
- 2 $(1, 8)$ not $(1, 10)$; only I, choose B

11 D

B+
Compound Interest

- 1 $A = 50000(1.02)^{24}$
- 2 ≈ 80422 , choose D

12 C

B
Ratios

- 1 $150x + 400y = 250x + 250y$
- 2 $x : y = 3 : 2$, choose C

13 A

A-
Variation

- 1 New $z = \frac{0.36}{1.2}z = 0.3z$
- 2 Decreased by 70%, choose A

14 B

B
Patterns and Sequences

- 1 $n\text{th} = 6 + (n - 1) \times 5$
- 2 9th = 46, choose B

15 D

B+
Surface Area of Solids

- 1 Slant height $= \sqrt{15^2 + 8^2} = 17$
- 2 TSA $= 256 + 544 = 800$, choose D

16 A

A
Ratios of Areas

- 1 $BE : EF : FC = 2 : 5 : 1$; use affine area ratios
- 2 $[CDYF] = 31 \text{ cm}^2$, choose A

17 C

A-
Triangles

- 1 $AB = BD$, $BC = CD$ give base angles
- 2 $\angle ACD = 40^\circ$, choose C

18 C

A
Similar Triangles

- 1 $AD = DE = 2EB$, $DF \parallel EC$, $CE = 64$
- 2 Similar triangles give $EF = 40$, choose C

19 B

A-
Quadrilaterals

- 1 $BD = \sqrt{30^2 - 18^2} = 24$
- 2 Trapezium area = 336 cm^2 , choose B

20 D

A-
Rhombuses

- 1 $BE = EF \Rightarrow \angle EBF = 62^\circ$
- 2 Rhombus $\Rightarrow \angle BDC = 62^\circ$, choose D

21 D

A-
Properties of Circles

- 1 arc $CD = 48^\circ$; $AC = BD$ gives equal arcs
- 2 $\angle ABD = 24^\circ$, choose D

22 B

A
Trigonometry

- 1 $AE = \frac{AB}{\tan \alpha}$
- 2 $\frac{AE}{ED} = \frac{1}{\tan \alpha \tan(\alpha + \beta)}$, choose B

23 A

A-
Equations of Straight Lines

- 1 From slope and intercept signs
- 2 II and III hold, choose A

24 A

A-
Equations of Straight Lines

- 1 Perpendicular: $\left(-\frac{3}{2}\right)\left(-\frac{k}{12}\right) = -1$
- 2 $k = -8$, choose A

25 C

B+
Transformations

- 1 $A(-5, -2)$ right 9: $B(4, -2)$
- 2 $C = (2, 4)$; y -coordinate 4, choose C

26 C

B+
Locus

- 1 Points at fixed distance 3 from a line
- 2 A pair of parallel lines, choose C

27 D

A-
Equations of Circles

- 1 Centre $(2, -3)$, $r = 4$
- 2 Area 16π ; centre in Q4; $(7, -3)$ outside; I and III, choose D

28 A

B+
Probability

- 1 $C_2^9 = 36$; 8 consecutive pairs
- 2 $P = \frac{2}{9}$, choose A

29 D

B+
Statistical Diagrams

- 1 Box-plot gives min, max, quartiles
- 2 Range and IQR obtainable, not SD; choose D

30 B

A-
Statistical Measures

- 1 Total 40; median at 20th–21st value
- 2 Median = 5, choose B

Section B

31 C

A-
Logarithmic Functions

- Intercepts 7, 8: $\frac{\log_2 y}{7} + \frac{\log_2 x}{8} = 1$
- $x^7 y^8 = 2^{56}$, choose C

32 A

A-
Logarithmic Equations

- Let $t = \log x$: $t^2 - t - 12 = 0$
- $t = 4$ or -3 , choose A

33 A

B+
Number Systems

- Convert **110101000110**₂
- $= 53 \times 2^6 + 6$, choose A

34 B

A-
Complex Numbers

- Multiply by $\frac{a+2i}{a+2i}$; denominator $a^2 + 4$
- Real part $= \frac{a}{a^2 + 4}$, choose B

35 D

A
Linear Programming

- Evaluate $5x + 4y + 7$ at vertices
- Max = **57**, choose D

36 A

A
Geometric Sequences

- $r^5 = 32 \Rightarrow r = 2$
- $a_{20} = 1048576$, choose A

37 B

A
Equations of Circles

- Centre $(-\frac{k}{10}, -\frac{2}{5})$
- $k = -152$, choose B

38 A

A
Sectors

- Sector $OAB = 60^\circ$; shaded = sector $OBC - \triangle$
- Shaded $\approx 11 \text{ cm}^2$, choose A

39 B

A
Circles and Tangents

- Tangent-chord $\angle EAT = 38^\circ$
- $\angle ADB = 56^\circ$, choose B

40 D

A+
Angles in 3-D

- $PQ = RQ \tan 47^\circ = SQ \tan 53^\circ$
- Gives $\angle RPS \approx 68^\circ$, choose D

41 D

A
Centres of Triangles

- Right angle at **B**
- Orthocentre at **B**, centroid inside; I and II, choose D

42 C

A
Permutations and Combinations

- At least 1 blue $= 1 - \frac{C_6^{11}}{C_6^{19}}$
- $= \frac{635}{646}$, choose C

43 B

A
Probability

- At most 2 correct $= 1 - \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{4}$
- $= \frac{7}{8}$, choose B

44 B

A-
Standard Scores

- Mary: $\bar{x} = 69 - 0.5(8) = 65$
- John: $65 - 12 = 53$, choose B

45 C

A-
Statistical Measures

- Each $\times 6$ then $+5$: mean $6m + 5$, range $6r$, variance $36v$
- All three true, choose C